Assignment 4 Code & Outputs

Case Study E

```python
### row of equations, while zdd is just the symbolic variable itself.
state_variable_form = sp.Matrix([[zd], [thetad], [zdd_eom], [thetadd_eom]])
states = sp.Matrix([[z], [theta], [zd], [thetad]])
inputs = sp.Matrix([[F]])


# finding the jacobian with respect to states (A) and inputs (B)
A = state_variable_form.jacobian(states)
B = state_variable_form.jacobian(inputs)

# sp.pprint(A)
# sp.pprint(B)

# sub in values for equilibrium points (x_e, u_e) or (x_0, u_0)
x_e = sp.Matrix([[sp.Rational(1, 2) * ell], [0], [0], [0]])
u_e = sp.Matrix([[sp.Rational(1, 2) * (m1+m2) * g]])
z_e = sp.symbols("z_e")
equilibrium = [(thetad, 0.0), (theta, 0.0), (zd, 0.0), (z, z_e), (F, g * (sp.Rational(1, 2) * m2 + m1 * z_e / ell))]
A_lin = sp.simplify(A.subs(equilibrium))
B_lin = sp.simplify(B.subs(equilibrium))

sp.pprint(A_lin)
sp.pprint(B_lin)

sp.pprint(sp.simplify(A_lin @ (states - x_e) + B_lin @ (inputs - u_e)))
```

A:

$$\begin{bmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0.0 & -g & 0 & 0 \\ \dfrac{-3\cdot g\cdot m_1}{ell^2\cdot m_2 + 3\cdot m_1\cdot z_e^2} & 0 & 0 & 0 \end{bmatrix}$$

B:

$$\begin{bmatrix} 0 \\ 0 \\ 0 \\ \dfrac{3\cdot ell}{ell^2\cdot m_2 + 3\cdot m_1\cdot z_e^2} \end{bmatrix}$$

Case Study F

```python
F, tau = sp.symbols("F tau")
force_mixing = [
    (fr, sp.Rational(1, 2) * F + sp.Rational(1,2) * tau / d),
    (fl, sp.Rational(1, 2) * F - sp.Rational(1,2) * tau / d)
    ]
state_variable_form = sp.Matrix([[zd], [hd], [thetad], [zdd_eom.subs(force_mixing)], [hdd_eom.subs(force_mixing)], [thetadd_eom.subs(force_mixing)]])
inputs = sp.Matrix([[F], [tau]])

# find equilibriums
equilibrium_vars = [z_v, h, theta, zd, hd, thetad, F, tau]
equilibrium_sol = sp.solve(state_variable_form, equilibrium_vars)

sp.pprint(equilibrium_sol)

# finding the jacobian with respect to states (A) and inputs (B)
A = state_variable_form.jacobian(state)
B = state_variable_form.jacobian(inputs)

# sp.pprint(A)
# sp.pprint(B)

# sub in values for equilibrium points (x_e, u_e) or (x_0, u_0)
equilibrium = [(z_v, 0.0), (h, 0.0), (theta, 0.0), (zd, 0.0), (hd, 0.0), (thetad, 0.0), (F, sp.symbols("F_e")), (tau, 0.0)]
A_lin = sp.simplify(A.subs(equilibrium))
B_lin = sp.simplify(B.subs(equilibrium))

sp.pprint(A_lin)
sp.pprint(B_lin)

sp.pprint(sp.simplify(A_lin @ state + B_lin @ inputs))
```

equilibria:

$$[(z_v(t), h(t), \pi, 0, 0, 0, -g\cdot(m_c + m_l + m_r), 0), (z_v(t), h(t), 0, 0, 0, 0, g\cdot(m_c + m_l + m_r), 0)]$$

A:

$$\begin{bmatrix} 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & \dfrac{-F_e}{m_c + 2\cdot m_r} & \dfrac{-\mu}{m_c + 2\cdot m_r} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

B:

$$\begin{bmatrix} 0 & 0 \\ 0 & 0 \\ 0 & 0 \\ 0 & 0 \\ \dfrac{1}{m_c + 2\cdot m_r} & 0 \\ 0 & \dfrac{1}{J_c + 2\cdot d^2\cdot m_r} \end{bmatrix}$$